DEPARTMENT OF APPLIED MATHEMATICS

Project 2 (PDE BVP) Report

VERIFIED FINITE-DIFFERENCE METHODS FOR MIXED BOUNDARY-VALUE PROBLEMS

Boundary Closures, Solver Accuracy, and Corner Regularity

GROUP E

Member	**Role**
Yi.Zhu (2361666)	Project Manager
Zhengdao.Song (2361511)	Mathematical Theory
Xiangyu.Zhang (2361653)	Algorithm Implementation Visualization and Report
Fangchi.Shi (2359837)	Algorithm Implementation
Rong.Yin (2363466)	Testing and Validation

October 2026

Abstract

This report derives and verifies finite-difference methods for the Poisson equation on the unit square with Dirichlet data on three sides and Neumann or Robin data on the open right edge. The study separates boundary consistency, algebraic error, and global solution error through manufactured data, exact-row substitution, mesh refinement, and independent solver and finite-element checks. Five-level experiments give global L_h^2 orders of 1.025404 and 1.020858 for one-sided Neumann and Robin closures, and 2.018947 and 2.022231 for ghost elimination. A tolerance sweep at $N = 128$ certifies rtol 10^{-10} as the first archived GMRES level with algebraic error below 10% of the discretisation error, while SuperLU is fastest at the tested common residual threshold. On an L-shaped domain, the P1 implementation gives (L^2, H^1) orders near $(2, 1)$ for smooth data and $(4/3, 2/3)$ for the $r^{2/3}$ singular solution. The resulting evidence supports ghost elimination on aligned square grids; extensions to coefficient jumps, thin boundary layers, and strongly distorted meshes remain open.

Keywords: mixed boundary conditions; ghost-point elimination; manufactured solutions; algebraic error; GMRES+ILU; corner singularity; P1 finite elements.

1 Introduction and Research Scope

1.1 Engineering Context and Research Questions

Mixed boundary conditions arise in steady heat conduction when selected walls are held at prescribed temperatures while another wall exchanges heat with its surroundings. The same elliptic model describes steady diffusion with an imposed flux or a convective exchange law. In each setting, the boundary row is part of the physical model: a low-order approximation of a flux or Robin law can dominate the global error even when the interior stencil is second order. We use this setting as a controlled proxy for boundary treatment in engineering simulation and ask three questions: how does the boundary closure determine the observed order, when is the algebraic solve accurate enough to expose that order, and how does reduced regularity alter the finite-element rate?

The Poisson equation with mixed boundary conditions provides a precise setting for studying the interaction between geometry, approximation, and linear algebra. Numerical accuracy couples the interior stencil, boundary closure, and algebraic solve. Exact substitution audits the assembled equations, mesh refinement measures discretisation error, and residual or backward-error measurements assess the algebraic solve. The objective is therefore a chain of evidence from the continuous model to the reported quantity of interest rather than a single convergence plot.

This study considers $-\Delta u = f$ on $\Omega = (0, 1)^2$, with Dirichlet data on the left, lower, and upper edges and a natural condition on the open right edge. Neumann and Robin data are generated from the same smooth solution, so closure effects are isolated under a common continuous problem. Two discrete treatments are derived in full. One-sided closure replaces the boundary PDE equation by a derivative relation; ghost-point elimination retains the PDE equation and removes an auxiliary exterior value. Both are standard constructions [1, 2]; their assembled rows differ in coefficients, units, local defects, and global error behaviour.

1.2 Verification Strategy and Scope

The methodology follows the established separation of code verification from solution verification and validation [3, 4]. Manufactured solutions make every forcing and boundary datum exactly computable [5]. The evidence chain has four layers: exact-row substitution checks signs and coefficients; five-level refinement

measures global and regional discretisation orders; a GMRES tolerance sweep separates algebraic from discretisation error; and matched P1-FEM, independent ordering, corner-regularity, and pure-Neumann tests provide independent controls. The accuracy budget is $\|x - x_h^\star\|_\infty < 0.1\|x_h^\star - u^*\|_\infty$, with $Ax_h^\star = b$, together with a scaled residual below 10^{-10}. The archived sweep stores scalar diagnostics rather than iterate vectors; a residual-to-error bound certifies this budget for the ghost Neumann matrix.

The core evidence comprises four systems: Neumann or Robin data combined with one-sided or ghost closure. Each is evaluated at $N = 16, 32, 64, 128, 256$ using global, interior, and right-edge error measures. Exact local defects are independently evaluated from Taylor formulas and compared with substitution into the assembled rows. Matrix nonsymmetry and row-scaled algebraic residuals are recorded separately. A second implementation with a different unknown ordering provides an assembly and indexing cross-check; both implementations use SciPy/SuperLU for the direct reference.

Two comparison studies extend the same verification logic. A separate pure-Dirichlet benchmark compares five-point FDM with P1 finite elements on identical physical nodes and degrees of freedom. On an L-shaped domain, one P1 implementation is applied to both a smooth exponential solution and the reentrant $r^{2/3}$ singular solution associated with nonsmooth elliptic theory [8]. Continuous L^2 and energy errors, an interpolation/discrete-error decomposition, and corner-region contributions distinguish regularity effects from implementation defects.

1.3 Principal Results and Report Structure

The one-sided Neumann and Robin closures approach first-order global accuracy, with last-three-level L_h^2 fits of 1.025404 and 1.020858. Ghost elimination gives corresponding fits of 2.018947 and 2.022231 and also approaches second order in the maximum norm. The algebraic results identify a practical GMRES threshold: rtol 10^{-8} remains visibly contaminated, rtol 10^{-10} reaches the discretisation plateau, and rtol 10^{-12} gives error norms within 0.02% of the direct-reference values. SuperLU gives the lowest total time at $N = 64, 128, 256$ because ILU setup dominates the short GMRES iteration phase.

The L-shaped control isolates the effect of regularity under a fixed mesh and solver protocol. Smooth P1 convergence is approximately second order in L^2 and first order in energy, whereas the singular case approaches $4/3$ and $2/3$. Section 2 derives the formulations and metrics, Section 3 gives the assembly and verification algorithms, Section 4 presents the numerical evidence and validity range, and Section 5 states the resulting design conclusions.

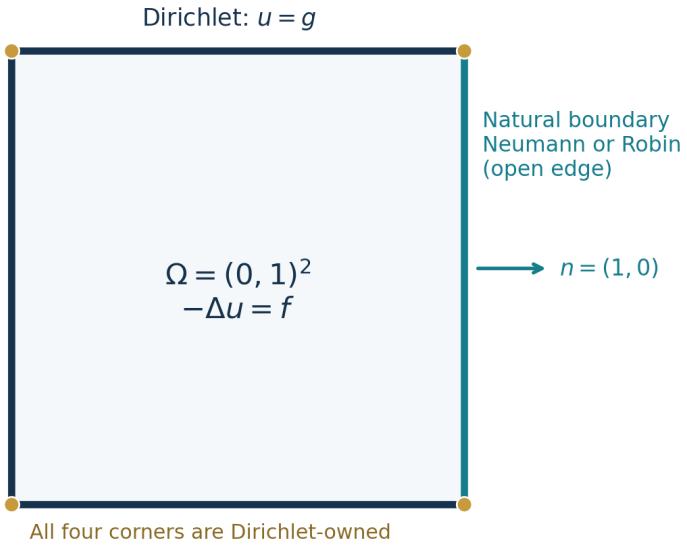


Figure 1. Boundary ownership in the unit-square model. The natural condition applies only on the open right edge; the outward normal there is $n = (1, 0)$.

2 Mathematical Model and Discrete Analysis

2.1 Continuous Model and Natural Boundary Terms

Let $\Omega = (0,1)^2$, with all grid corners assigned to the Dirichlet set. The two models and their boundary ownership are

$$\begin{gathered} -\Delta u = f \text{ in } \Omega, \qquad u = g \text{ on } \Gamma_D, \qquad \beta = 2, \\ \begin{cases} \partial_n u = q, & \text{Neumann,} \\ \partial_n u + \beta u = r, & \text{Robin,} \end{cases} \quad \text{on } \Gamma_R, \\ \Gamma_D = (\{0\} \times [0,1]) \cup ([0,1] \times \{0,1\}), \qquad \Gamma_R = \{1\} \times (0,1). \end{gathered} \tag{1}$$

The manufactured solution and all required data are

$$\begin{gathered} u^* = e^{x+y}, \qquad f = -\Delta u^* = -2e^{x+y}, \qquad g = u^*|_{\Gamma_D}, \\ q(y) = u_x^*(1,y) = e^{1+y}, \qquad r(y) = q(y) + \beta u^*(1,y) = 3e^{1+y}. \end{gathered} \tag{2}$$

Both natural-boundary problems therefore share the same exact solution.

For $V_0 = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}$, integration by parts gives the two weak forms

$$\begin{gathered} \int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx + \int_{\Gamma_R} q v \, ds, \\ \int_\Omega \nabla u \cdot \nabla v \, dx + \beta \int_{\Gamma_R} u v \, ds = \int_\Omega f v \, dx + \int_{\Gamma_R} r v \, ds. \end{gathered} \tag{3}$$

Coercivity follows from the nonempty Dirichlet boundary, with $\beta > 0$ strengthening the Robin case.

2.2 Interior Stencil and Local Defect

The uniform grid, ordering, and unknown count are

$$\begin{gathered} h = \frac{1}{N}, \quad x_i = ih, \quad y_j = jh, \quad k = (j-1)N + (i-1), \\ 1 \le i \le N, \qquad 1 \le j \le N-1, \qquad m = N(N-1). \end{gathered} \tag{4}$$

Right-edge interior nodes remain unknowns; Dirichlet nodes are eliminated.

Taylor expansion gives the centred second difference

$$u_{i+1,j} - 2u_{i,j} + u_{i-1,j} = h^2 u_{xx} + \frac{h^4}{12} u_{xxxx} + O(h^6). \tag{5}$$

The interior five-point equation and its exact local defect are therefore

$$\begin{aligned} \frac{4U_{i,j} - U_{i-1,j} - U_{i+1,j} - U_{i,j-1} - U_{i,j+1}}{h^2} &= f_{i,j}, \\ (Au^* - b)_{i,j} &= -\frac{h^2}{12}(u^*_{xxxx} + u^*_{yyyy}) + O(h^4) \\ &= -\frac{h^2}{6} e^{x_i + y_j} + O(h^4). \end{aligned} \tag{6}$$

For free values u_I and known Dirichlet values g_D, elimination gives

$$\begin{gathered} A_{II} u_I = b_I - A_{ID} g_D, \\ b_k \longleftarrow b_k - a_{k\ell} g_\ell. \end{gathered} \tag{7}$$

For $a_{k\ell} = -1/h^2$, the scalar update is $b_k \leftarrow b_k + g_\ell/h^2$.

2.3 One-Sided and Ghost Boundary Closures

At the right edge, one-sided closure replaces the boundary PDE row by

$$\begin{cases} \dfrac{U_{N,j} - U_{N-1,j}}{h} = q_j, & \text{Neumann,} \\ \left(\dfrac{1}{h} + \beta\right) U_{N,j} - \dfrac{1}{h} U_{N-1,j} = r_j, & \text{Robin.} \end{cases} \tag{8}$$

The backward-difference expansion gives the shared defect

$$\begin{aligned} D_- u^* - u^*_x &= -\frac{h}{2} u^*_{xx} + O(h^2), \\ \tau^{(1)}_{N,j} &= -\frac{h}{2} u^*_{xx}(1, y_j) + O(h^2). \end{aligned} \tag{9}$$

Ghost elimination retains the boundary PDE row and uses

$$\begin{aligned} D_0 u^* &= u^*_x + \frac{h^2}{6} u^*_{xxx} + O(h^4), \\ U_{N+1,j} &= \begin{cases} U_{N-1,j} + 2h q_j, & \text{Neumann,} \\ U_{N-1,j} + 2h(r_j - \beta U_{N,j}), & \text{Robin.} \end{cases} \end{aligned} \tag{10}$$

Substitution gives the complete solver rows

$$\begin{aligned} \frac{4U_{N,j} - 2U_{N-1,j} - U_{N,j-1} - U_{N,j+1}}{h^2} &= f_{N,j} + \frac{2q_j}{h}, \\ \frac{(4 + 2\beta h)U_{N,j} - 2U_{N-1,j} - U_{N,j-1} - U_{N,j+1}}{h^2} &= f_{N,j} + \frac{2r_j}{h}. \end{aligned} \tag{11}$$

Table 1. Nonzeros and $\delta_A = \|A - A^T\|_F / \|A\|_F$ for the four mixed systems. The two nnz columns are common to Neumann and Robin at each closure.

N	nnz one-sided	δ_A N, 1s	δ_A R, 1s	nnz ghost	δ_A N, g	δ_A R, g
16	1110	0.076931	0.076931	1138	0.079197	0.078942
32	4774	0.055155	0.055155	4834	0.055948	0.055904
64	19782	0.039265	0.039265	19906	0.039544	0.039537
128	80518	0.027858	0.027858	80770	0.027956	0.027955
256	324870	0.019731	0.019731	325378	0.019766	0.019766

Elimination changes the row being measured. For either natural condition, the complete-row defect is

$$\begin{aligned} \tau^{(g)}_{N,j} &= (Au^* - b)_{N,j} \\ &= -\frac{h^2}{12}(u^*_{xxxx} + u^*_{yyyy}) + \frac{2}{h}(D_0 u^* - \partial_n u^*) + O(h^2) \\ &= \frac{h}{3} u^*_{xxx}(1, y_j) + O(h^2). \end{aligned} \tag{12}$$

The row defect is therefore $O(h)$ even though the interface derivative is second order.

Global rates follow from the defect distribution and the inverse discrete operator. For $e = U - u^*$ and $\tau = Au^* - b$,

$$Ae = -\tau, \qquad e = -A^{-1}\tau, \qquad \|e\| \le \|A^{-1}\| \, \|\tau\|. \tag{13}$$

The boundary rows are few and differently scaled, while A^{-1} distributes their influence. Global accuracy must therefore be measured directly.

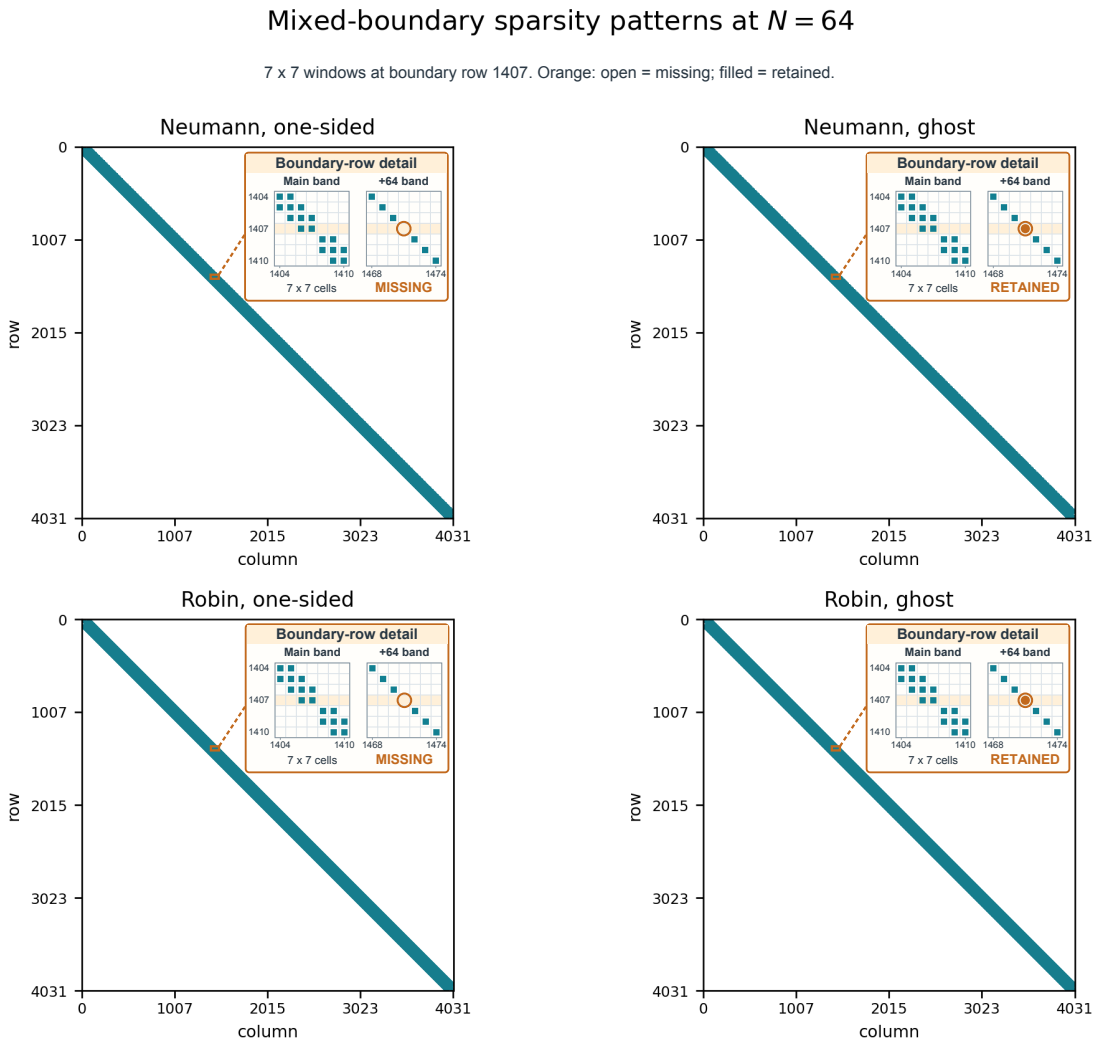


Figure 2. Four mixed-boundary sparsity patterns at $N = 64$. Neumann and Robin data share the same graph for a fixed closure; ghost elimination adds the vertical couplings associated with the retained boundary PDE row.

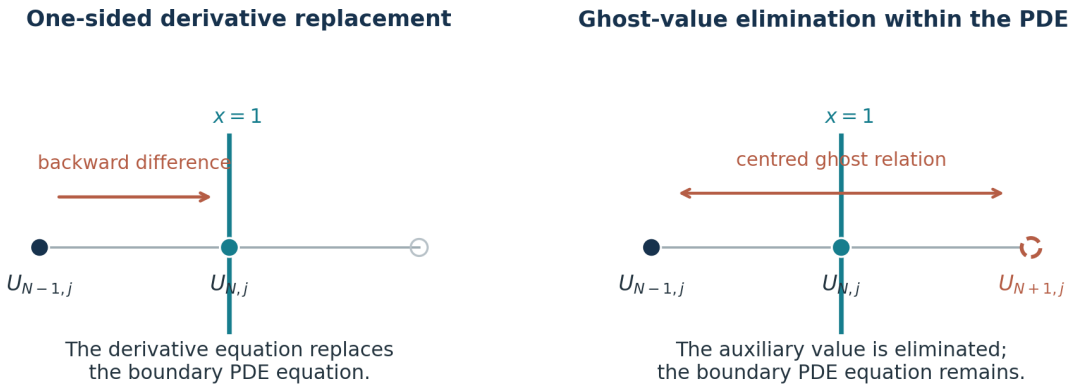


Figure 3. The two closures are algebraically distinct row constructions. One-sided closure replaces the PDE equation; ghost elimination modifies its inward coupling and load.

2.4 Error Norms and Algebraic Error

On a node set S, let $e_k = U_k - u^*(x_k, y_k)$. The discrete error measures are

$$E_{2,h}(S) = \left(h^2 \sum_{k \in S} e_k^2 \right)^{1/2}, \qquad E_\infty(S) = \max_{k \in S} |e_k|,$$
$$E_{2,\Gamma} = \left(h \sum_{k \in \Gamma_R} e_k^2 \right)^{1/2}. \tag{14}$$

Consecutive-grid orders are

$$p_k = \frac{\log(E_{k-1}/E_k)}{\log(N_k/N_{k-1})}. \tag{15}$$

The reported positive last-three-level order is the negative least-squares slope of $\log E_k$ against $\log N_k$ (equivalently, the slope against $\log h_k$).

For the exact discrete solution $x_h^\star$ and an iterative vector x, the error split is

$$x - u^* = (x - x_h^\star) + (x_h^\star - u^*) = \delta_{\text{alg}} + e_{\text{disc}},$$
$$R_{\text{alg}} = \frac{\|x - x_h^\star\|_\infty}{\|x_h^\star - u^*\|_\infty}. \tag{16}$$

The computed direct vector x_d supplies a numerical reference. The validation protocol uses maximum-entry row scaling; the row-sum scaling in the benchmark audit is retained as an independent check:

$$(D_{\max})_{ii} = \max_j |A_{ij}|, \qquad \rho_{\max} = \|D_{\max}^{-1}(b - Ax)\|_\infty. \tag{17}$$

For ghost Neumann closure, $D_{\max} = (4/h^2)I$. The reduced matrix has nonpositive off-diagonal entries, weakly diagonally dominant rows, and a path from every row to a strictly dominant row beside Γ_D; hence it is a nonsingular M-matrix with $A^{-1} \geq 0$. The grid function $w(x, y) = x - x^2/2$ satisfies the full ghost stencil $L_h w = 1$. Its nonnegative eliminated Dirichlet values imply $Aw_I \geq \mathbf{1}$, with $0 \leq w_I \leq 1/2$. Thus, for $d = x - x_h^\star$,

$$\|d\|_\infty \leq \tfrac{1}{2}\|b - Ax\|_\infty = 2N^2 \rho_{\max} =: B,$$
$$R_{\text{alg}} \leq R_{\text{cert}} := \frac{B}{E_\infty(x) - B}, \qquad E_\infty(x) > B.$$

The last bound uses $\|x_h^\star - u^*\|_\infty \geq E_\infty(x) - B$, so archived residuals and total errors suffice for certification.

Matrix structure is measured by

$$\delta_A = \frac{\|A - A^T\|_F}{\|A\|_F}. \tag{18}$$

For $\delta_A > 0$, conjugate gradients is assessed experimentally; the standard SPD analysis and the observed nonsymmetric convergence are reported separately for each finite-precision run.

2.5 P1 FEM and Regularity-Dependent Error

On a triangular mesh $\mathcal{T}_h$, let $V_h \subset H^1$ be continuous and piecewise affine. The pure-Dirichlet system is

$$K_{ij} = \int_\Omega \nabla \phi_j \cdot \nabla \phi_i \, \mathrm{d}x, \quad b_i = \int_\Omega f \phi_i \, \mathrm{d}x,$$
$$K_{II} u_I = b_I - K_{ID} g_D. \tag{19}$$

On the fixed-diagonal right-isosceles triangulation, the basis gradients opposite a hypotenuse are orthogonal, so their stiffness coupling vanishes. With A_{FDM} storing the $1/h^2$-scaled five-point operator, the common-

node identity is

$$h^2 A_{\mathrm{FDM}} = K_{\mathrm{P1}} \qquad \Longleftrightarrow \qquad A_{\mathrm{FDM}} - N^2 K_{\mathrm{P1}} = 0, \tag{20}$$

whereas $b_{\mathrm{P1}} \neq h^2 b_{\mathrm{FDM}}$ in general because integrated load moments replace nodal samples.

Continuous FEM errors are evaluated as

$$\|e\|^2_{L^2(\Omega)} = \sum_{T \in \mathcal{T}_h} \int_T e^2 \,\mathrm{d}x, \qquad |e|^2_{H^1(\Omega)} = \sum_{T \in \mathcal{T}_h} \int_T |\nabla e|^2 \,\mathrm{d}x. \tag{21}$$

Galerkin orthogonality underlies the following regularity argument.

On the L-shaped domain $\Omega_L = (-1, 1)^2 \setminus [0, 1)^2$, the singular control is

$$u^* = r^\alpha \sin(\alpha\phi), \qquad \alpha = \frac{2}{3}, \qquad 0 \le \phi \le \frac{3\pi}{2}. \tag{22}$$

The angle ϕ is measured counterclockwise from the positive y-axis through Ω_L. This harmonic control vanishes on the cut edges and satisfies

$$|\nabla u^*| \sim C r^{\alpha - 1} = C r^{-1/3}, \qquad u^* \in H^{1+s}(\Omega_L) \text{ only for } s < \alpha = \frac{2}{3}. \tag{23}$$

With nodal Dirichlet data $g_h = I_h g$, let $e = u_h - u^*$ and $-\Delta z = e$, $z|_{\partial\Omega_L} = 0$. The dual shift gives $\|z\|_{H^{1+s}} \le C_s \|e\|_{L^2}$ only for $s < 2/3$. Green's identity and Galerkin orthogonality yield

$$\begin{aligned} \|e\|^2_{L^2} &= a(e, z - I_h z) - \langle g_h - g, \partial_n z \rangle_{\partial\Omega_L}, \\ \|e\|_{L^2} &\le C_s h^s |e|_{H^1} + C \|g_h - g\|_{H^{1/2}(\partial\Omega_L)}, \quad 0 < s < \tfrac{2}{3}. \end{aligned} \tag{24}$$

The normal derivative is the weak trace. The boundary norm is $O(h^{3/2})$ because g is smooth edgewise and zero on the cut edges. Together with $|e|_{H^1} = O(h^{2/3-\eta})$ for $\eta > 0$, this proves $\|e\|_{L^2} = O(h^{4/3-\varepsilon})$ for every $\varepsilon > 0$. The observed $h^{4/3}$ rate is consistent with this estimate; the endpoint shift is not assumed [8, 9].

For the smooth leg, the nodal interpolant gives

$$e = u_h - u^* = (u_h - u_I) - (u^* - u_I) = \theta - \rho. \tag{25}$$

The interpolation component satisfies $\|\rho\|_{L^2} = O(h^2)$ and $|\rho|_{H^1} = O(h)$; the measured order of θ determines whether the total smooth L^2 error remains second order.

Quadrature must preserve these rates. On the reference triangle $\widehat{T}$, the Duffy transformation gives

$$\begin{aligned} (\xi, \eta) &= (s, (1 - s)t), \\ \int_{\widehat{T}} F \,\mathrm{d}\xi \,\mathrm{d}\eta &= \int_0^1 \int_0^1 F(s, (1 - s)t)(1 - s) \,\mathrm{d}s \,\mathrm{d}t. \end{aligned} \tag{26}$$

An affine map to each physical triangle contributes the Jacobian factor $|\det J_T|$. Tensor Gauss rules are used away from the corner. Corner elements use the analytic radial factor

$$\int_0^R r^a \,\mathrm{d}r = \frac{R^{a+1}}{a+1} \qquad (a > -1), \tag{27}$$

with adaptive angular quadrature and singular-gradient nodes restricted to $r > 0$.

2.6 Pure-Neumann Compatibility and Gauge

Pure Neumann data require load compatibility, while constants remain undetermined. With discrete load defect $d = \mathbf{1}^T b$ and mass matrix M, the two conditions and compatible correction are

$$\begin{gathered} \int_\Omega f \,\mathrm{d}x + \int_{\partial\Omega} q \,\mathrm{d}s = 0, \\ b_c = b - \frac{d}{\mathbf{1}^T M \mathbf{1}} M\mathbf{1}, \qquad \mathbf{1}^T b_c = 0. \end{gathered} \tag{28}$$

The remaining constant mode requires a gauge. The mean-gauged saddle-point system

$$Ku + M\mathbf{1}\,\lambda = b_c, \qquad (M\mathbf{1})^T u = \int_\Omega u^* \,\mathrm{d}x = (e-1)^2 \tag{29}$$

selects the intended solution while preserving every equation.

3 Implementation and Verification Design

3.1 Modular Assembly and Reproducibility

The implementation separates exact data, grid ownership, sparse assembly, linear solvers, and error measurement. The assembler receives N, a boundary variant, and a closure; it constructs the coordinate map with right corners absent, adds coordinate triplets, applies Equation (7), and converts the system to CSR form. Row labels distinguish interior PDE equations from right-edge closure equations. Exact-row substitution evaluates $Au^* - b$ before the solve, whereas post-solve diagnostics evaluate $b - Ax$ and $u_h - u^*$.

The reference stack uses Python 3.12.12 with NumPy 2.2.5, SciPy 1.16.2, Matplotlib 3.10.3, and scikit-fem 12.0.2. SuperLU is the direct reference for the sparse nonsymmetric systems [7]; GMRES and GMRES+ILU provide iterative comparisons for the same matrix and right-hand side [6]. The direct method is selected for the refinement tables because it removes iteration tolerance from the discretisation-rate estimate. GMRES is the general iterative method for the mixed nonsymmetric matrix, while CG, Jacobi, Gauss–Seidel, and SOR are retained as measured comparison points. The supplied archive contains recorded CSV data and report sources; neither `code/run_all.py` nor a `validation/` directory was supplied. Independent corner-quadrature and right-row checks, followed by figure generation, are reproduced from the archive root with

```
python corrections/recompute_corner_and_right_rows.py
        python corrections/build_figures.py
```

These commands reproduce the added independent checks and figures from archived data. The full solver study has not been rerun; solver timings remain the archived measurements.

Algorithm 1: Mixed-boundary sparse assembly

Input: N, boundary type (Neumann or Robin), closure (one-sided or ghost), f, g, q, r, β.

1. Set $h = 1/N$ and map only free nodes (i, j) to vector indices; assign all corners to Γ_D.
2. For each interior free node, insert the five-point coefficients in COO form and move each known Dirichlet term to the right-hand side using Equation (7).
3. For each right-edge node, insert either the one-sided row in Equation (8) or the ghost-eliminated PDE row in Equation (11).
4. Convert the triplets to CSR form, retain row labels, and return (A, b).

Stationary iterations are implemented directly. Jacobi uses a simultaneous diagonal correction; Gauss–Seidel is SOR with $\omega = 1$; SOR reuses one sparse triangular factorisation of $D + \omega L$. Every iterate is assessed by a freshly computed residual. ILU setup and GMRES iteration are timed separately before their costs are added.

3.2 Verification Protocols

Core convergence tables use direct solves to isolate the reported discretisation orders from iterative contamination. The GMRES tolerance sweep fixes $N = 128$, ghost Neumann closure, and the exact solution while varying only rtol. The full solver ladder uses the same assembler and a common stop $\rho_{\max} \le 10^{-10}$. Wall times are single-run measurements in the recorded environment; iteration counts, residuals, and deterministic errors are the portable evidence.

Algorithm 2: Refinement and algebraic-error audit

Input: $N \in \{16, 32, 64, 128, 256\}$ and GMRES tolerances $10^{-2}, \ldots, 10^{-12}$.
1. Assemble and solve each mixed system with SuperLU; compute $E_{2,h}$, E_∞, the right-edge norm, $\rho_{\max}$, and $\tau = Au^* - b$.
2. Fit the last three values of $\log E$ against $\log N$ and store the negative slope as the positive observed order.
3. At $N = 128$, read the archived CSV scalars. Use rounding endpoints ρ^+ (upper residual) and E_∞^- (lower observed error); set $B^+ = 2N^2\rho^+$.
4. If $E_\infty^- > B^+$, set $R_{\text{cert}} = B^+/(E_\infty^- - B^+)$. Accept when $\rho^+ \le 10^{-10}$ and $R_{\text{cert}} < 0.1$; the first certified archived level is rtol 10^{-10}.

The L-shaped study uses identical meshes, P1 assembly, continuous norms, and error integration for its smooth and singular legs. Mesh area is exactly three at every level, and the re-entrant corner is Dirichlet-owned. Corner elements use analytic radial integration with adaptive angular quadrature; other elements use Duffy tensor-Gauss integration of order eight for L^2. The piecewise-affine $\theta = u_h - u_I$ is integrated exactly by a three-point edge-midpoint rule because θ^2 is quadratic on each triangle.

Algorithm 3: P1 regularity control

Input: mesh family and refinement levels; smooth and singular exact solutions.
1. Generate the mesh and mark the boundary, including the re-entrant corner, as Dirichlet-owned.
2. Assemble K and F, eliminate $K_{ID}g_D$, and solve the reduced system directly.
3. Integrate continuous L^2 and H^1 errors using corner-exact radial/angular quadrature and Duffy quadrature elsewhere.
4. Report errors, quadrature self-checks, corner-region shares, and fitted rates against h and actual DOFs.

Table 2. Verification layers used in the final evidence set. Each layer answers a different question and supplies complementary evidence.

Evidence layer	Fixed protocol	Quantity established
Manufactured data	Exact differentiation and boundary substitution	Signs, coefficients, fluxes, Robin datum
Complete-row audit	$Au^* - b$ on independent formulas	Interior and boundary Taylor defects
Five-level refinement	$N = 16, 32, 64, 128, 256$, direct solves	Global and regional discretisation orders
Algebraic audit	$D_{\max}$ scaling, residual and certified forward bound	Solver accuracy separate from model accuracy
Matched P1 controls	Same meshes, norms, integration and solver	Regularity, interpolation, and corner effects

The pure-Dirichlet FDM/P1-FEM benchmark sorts coordinates by algebraic unknown number and asserts equality before comparing errors. The structured-grading comparison is presented in Section 4.8.

3.3 Verification Controls and Validity Range

Substitution tests enforce the Dirichlet sign convention and the complete Robin load. Common-node assertions establish a shared ordering before FDM/P1 comparison. Corner ownership and corner-exact quadrature define the L-shaped error functional, while independent solver ordering and the pure-Neumann compat-

ibility and gauge tests provide separate checks on nullspace handling. These controls make the local defect $\tau = Au^* - b$, algebraic residual $b - Ax$, and global solution error $u_h - u^*$ distinct reported quantities.

The conclusions apply to aligned square grids, the smooth manufactured data used for the mixed-boundary study, and the tested range $N \leq 256$. Timing values are single-run measurements in the stated environment. CG convergence and the SOR parameter are empirical observations for the mildly nonsymmetric matrix family; the general iterative recommendation is GMRES. Coefficient jumps, thin boundary layers, strongly distorted meshes, and estimator-driven adaptivity require additional experiments beyond the present protocol.

4 Numerical Results and Verification

4.1 Mixed-Boundary Convergence

The four mixed systems are solved directly at $N = 16, 32, 64, 128, 256$. Figure 4 and Table 3 use the same global area-weighted norm throughout. One-sided Neumann errors have successive L_h^2 orders of 1.1312, 1.0670, 1.0338, and 1.0170; ghost Neumann errors give 2.0861, 2.0478, 2.0251, and 2.0128. The corresponding one-sided Robin sequence is 1.1054, 1.0547, 1.0278, and 1.0140, while ghost Robin gives 2.1012, 2.0561, 2.0294, and 2.0150. The five-level refinement sequence establishes the asymptotic distinction between the two closure families.

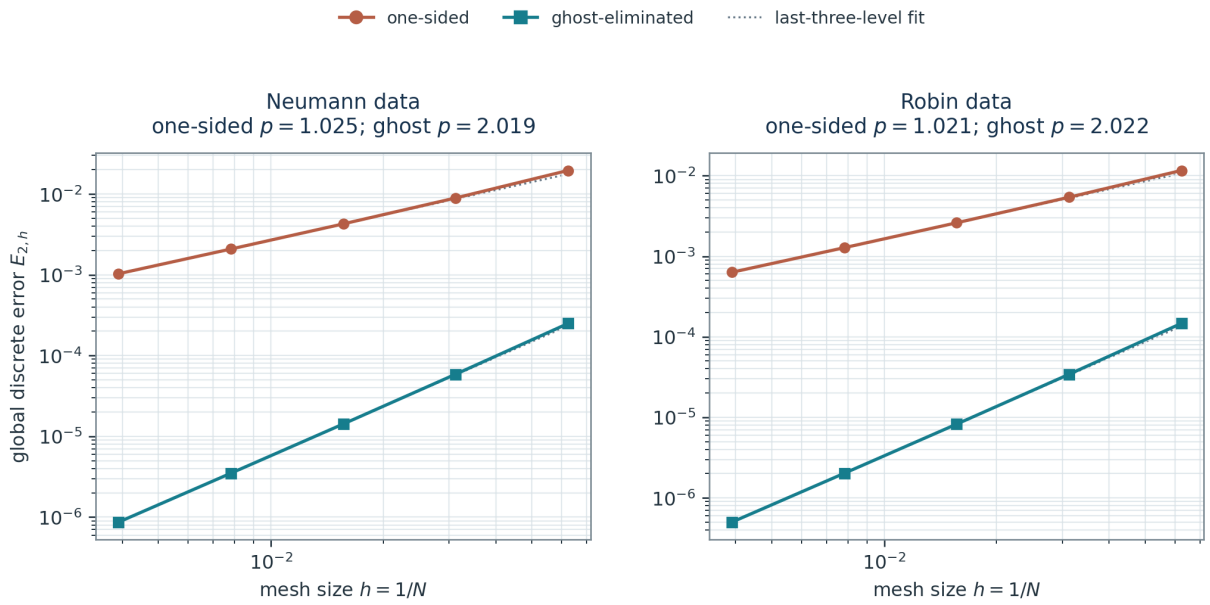


Figure 4. Five-level global L_h^2 convergence for all mixed-boundary systems. Dotted fits use $N = 64, 128, 256$; their positive observed orders are reported in the panel titles so that the data region remains unobstructed.

Table 3. Global discrete L_h^2 errors on the five core meshes. Every entry comes from a direct sparse solve of the indicated assembled system.

N	Neumann one-sided	Neumann ghost	Robin one-sided	Robin ghost
16	1.9578×10^{-2}	2.4976×10^{-4}	1.1623×10^{-2}	1.4669×10^{-4}
32	8.9385×10^{-3}	5.8824×10^{-5}	5.4021×10^{-3}	3.4188×10^{-5}
64	4.2663×10^{-3}	1.4227×10^{-5}	2.6005×10^{-3}	8.2209×10^{-6}
128	2.0837×10^{-3}	3.4955×10^{-6}	1.2755×10^{-3}	2.0138×10^{-6}
256	1.0297×10^{-3}	8.6614×10^{-7}	6.3161×10^{-4}	4.9822×10^{-7}

Table 4 adds maximum-norm and regional fits over the three finest levels. The one-sided global fits are close to one in every norm, whereas the ghost fits are close to two. The right-edge line norm is especially direct evidence of improved interface accuracy, and the interior fits show that its influence extends into the

domain. At $N = 256$, ghost Neumann gives $E_\infty = 3.3845 \times 10^{-6}$, compared with 3.4109×10^{-3} for one-sided closure; the corresponding Robin values are 2.0851×10^{-6} and 2.0745×10^{-3}.

Table 4. Last-three-level fits over $N = 64, 128, 256$. The interior norm is weighted by h^2 and the right-edge line norm by h.

Case	Global L_h^2	R^2	Global L_∞	Interior L_h^2	Edge line
Neumann, one-sided	1.025404	0.999978	1.009782	0.997743	1.011825
Neumann, ghost	2.018947	0.999997	1.999631	1.979608	1.999495
Robin, one-sided	1.020858	0.999985	1.004940	0.992763	1.007212
Robin, ghost	2.022231	0.999996	1.999775	1.976125	1.999440

The smaller Robin ghost error constant is specific to this exponential solution and $\beta = 2$, while the closure-based order comparison remains unchanged. Across all four finest-level systems, the direct-solve scaled residuals remain between 5.77×10^{-15} and 8.55×10^{-15} using the $D_{\max}$ scaling in Equation (17). These measurements attribute the observed error differences to the discretisation closures.

4.2 Local Defects and Global Error

An independent row audit evaluates $|(Au^* - b)_{N,j}|$ at the common grid node $y_j = 3/8$, with $j = 3N/8$ an integer on every listed mesh. Neumann and Robin values coincide here: for the one-sided rows the Robin contribution cancels in $r - q - \beta u^* = 0$, and the same cancellation occurs after ghost elimination. Table 5 shows halving defects and observed orders approaching one for both constructions.

Table 5. Absolute complete-row defects at $y = 3/8$. The order p_k is computed between consecutive rows; Neumann and Robin audits give the same tabulated values for this manufactured solution.

N	One-sided defect	Order	Ghost-row defect	Order
16	$1.210\,610 \times 10^{-1}$	—	$7.983\,827 \times 10^{-2}$	—
32	$6.115\,934 \times 10^{-2}$	0.9851	$4.055\,698 \times 10^{-2}$	0.9771
64	$3.073\,873 \times 10^{-2}$	0.9925	$2.043\,868 \times 10^{-2}$	0.9886
128	$1.540\,936 \times 10^{-2}$	0.9963	$1.025\,948 \times 10^{-2}$	0.9943

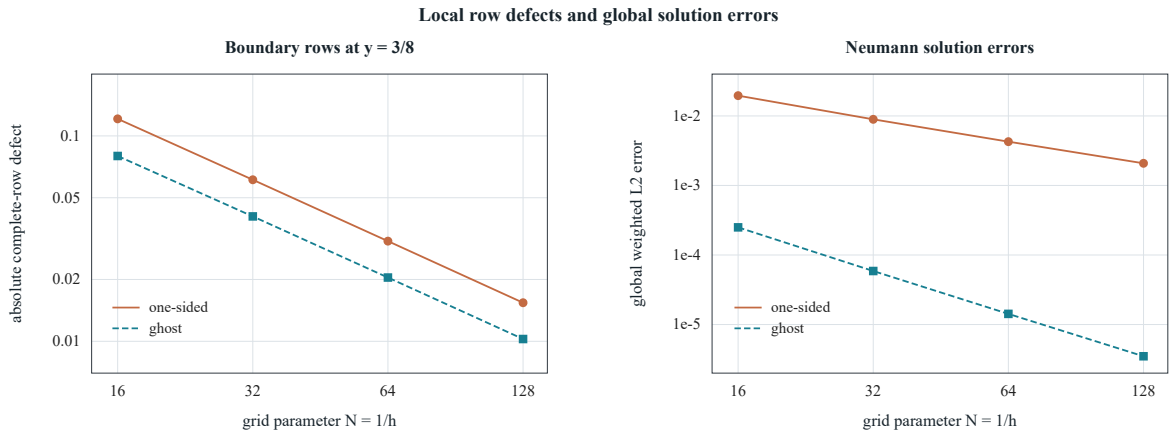


Figure 5. Local complete-row defects and global Neumann errors measure different objects. Both defect families are locally $O(h)$ at $y = 3/8$, while the global ghost error follows an approximately second-order trend.

This result is consistent with Equations (9), (12), and (13). The centred derivative relation used to construct the ghost value is second order, and the eliminated PDE row retains an unscaled $O(h)$ defect. The global solution error follows the separate relation $e = -A^{-1}\tau$.

4.3 GMRES Tolerance and the Error Plateau

For the ghost Neumann system at $N = 128$, the direct reference is $E_{2,h} = 3.4955 \times 10^{-6}$ and $E_\infty = 1.3537 \times 10^{-5}$. Table 6 varies only the unpreconditioned GMRES rtol. Tightening from 10^{-2} to 10^{-12} reduces the scaled residual monotonically, while the solution error approaches its discretisation-error plateau.

Table 6. Unpreconditioned GMRES tolerance sweep for ghost Neumann closure at $N = 128$. The residual uses Equation (17); the last column is a rounding-safe upper bound on R_{alg}, not a measured iterate difference. A dash means the bound is inconclusive.

rtol	Scaled residual	L_h^2 error	L_∞ error	R_{cert} bound
1×10^{-2}	5.27×10^{-3}	2.6253	5.3531	—
1×10^{-4}	2.78×10^{-5}	6.3702×10^{-2}	1.2668×10^{-1}	—
1×10^{-6}	3.40×10^{-7}	6.2350×10^{-4}	1.2480×10^{-3}	—
1×10^{-8}	3.76×10^{-9}	8.3410×10^{-6}	2.3784×10^{-5}	—
1×10^{-10}	2.99×10^{-11}	3.5447×10^{-6}	1.3654×10^{-5}	≤ 0.0775
1×10^{-12}	2.99×10^{-13}	3.4960×10^{-6}	1.3538×10^{-5}	≤ 0.000726

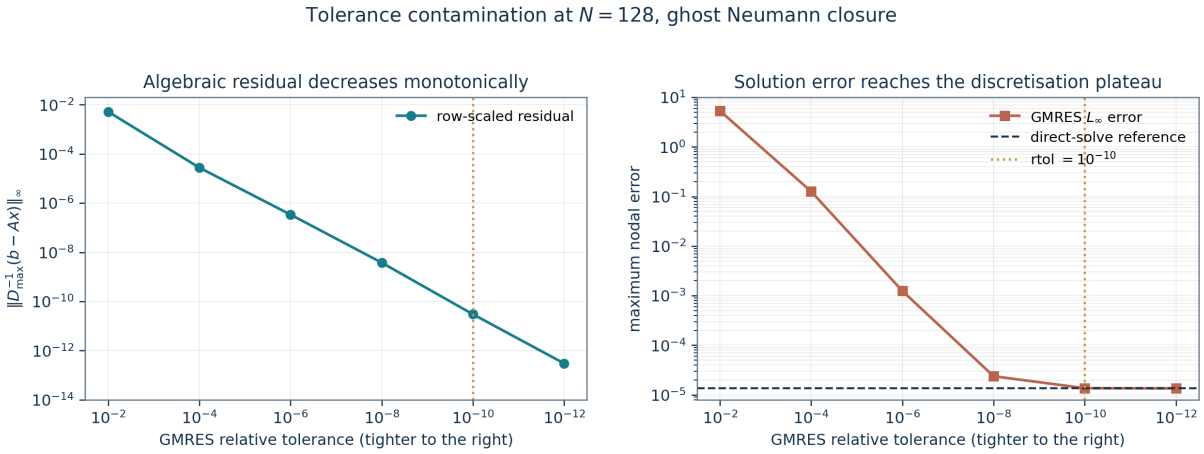


Figure 6. Residual tightening and the emergence of the discretisation-error plateau at $N = 128$. The error panel includes the direct reference; residual and error axes use independent scales.

At rtol 10^{-8}, the maximum error is 1.757 times the direct reference and the L_h^2 error is 2.386 times that reference, together with a scaled residual of 3.76×10^{-9}. At 10^{-10}, these ratios fall to 1.0086 and 1.0141: the run has entered the plateau and remains about 0.9% above the direct maximum error. At 10^{-12}, both error norms agree with the direct-reference errors to within 0.02%. To certify the algebraic-error budget from the archived scalars, let ρ^+ and E_∞^- be the upper and lower rounding endpoints of the residual and observed maximum error. Set $B^+ = 2N^2\rho^+$ and use $R_{\text{cert}} = B^+/(E_\infty^- - B^+)$ when the denominator is positive. Assuming nearest rounding to the displayed precision gives $R_{\text{cert}} = 0.077446 < 0.1$ at rtol 10^{-10} and 0.00072548 at 10^{-12}. Earlier levels do not certify the budget with this bound. These are residual-based bounds, not recovered iteration vectors.

4.4 Solver Iterations and Total Cost

The full ladder uses ghost Neumann systems and the common stop $\rho_{\max} \le 10^{-10}$. Table 7 reports wall seconds and iteration counts from the same recorded run. Jacobi and Gauss–Seidel are omitted at $N = 256$ after their measured growth through $N = 128$; the other methods are run at all three levels.

The last-row bounds exceed the $0.1E_{\text{disc},\infty}$ budget at $N = 128$ and 256, so this ladder compares cost at equal residual threshold rather than certified equal solution accuracy. At that common stop, sparse SuperLU records the lowest total time at every tested size. GMRES+ILU is the closest iterative method, with total times 1.4, 1.9, 1.8 times the direct times at $N = 64, 128, 256$. Its ILU setup costs 0.012, 0.096, 0.385 s, followed by 0.003, 0.013, 0.150 s of GMRES iteration; the small iteration counts coexist with a sub-

Table 7. Mixed-system solver ladder. Each cell gives wall time in seconds followed by iterations; a dash denotes a direct solve or an omitted run. SOR uses the measured value $\omega = 1.900$. The last row bounds the absolute algebraic maximum error at the common residual stop; it is not a timing row.

Method / bound	$N = 64$: time / iter.	$N = 128$: time / iter.	$N = 256$: time / iter.
SuperLU	0.010 / —	0.056 / —	0.305 / —
Jacobi	1.262 / 23250	28.113 / 85630	— / —
Gauss–Seidel	2.511 / 11620	57.296 / 42810	— / —
SOR	0.139 / 590	3.290 / 2480	62.367 / 9360
CG	0.014 / 425	0.210 / 748	1.873 / 1351
GMRES+ILU	0.014 / 5	0.108 / 6	0.535 / 9
$B_\infty = 2N^2\,10^{-10}$	8.192×10^{-7}	3.2768×10^{-6}	$1.310\,72 \times 10^{-5}$

stantial construction cost. Separately, unpreconditioned GMRES at the tighter matched rtol 10^{-12} takes $0.141, 5.194, 47.865$ s, compared with direct times $0.009, 0.053, 0.302$ s in that experiment.

The iteration measurements distinguish theoretical predictions from observed mixed-system behaviour. On the pure-Dirichlet model, the theoretical SOR values $\omega_{\text{opt}} = 1.906455$ and 1.952093 require 270 and 530 iterations at $N = 64, 128$; a 0.025-spaced sweep at $N = 64$ selects 1.900 with 310 iterations. For the mixed system, the sweep selects 1.900 with 590 iterations, providing an empirical optimum for the assembled matrix. On the SPD Dirichlet matrix, CG takes 436 iterations at $N = 128$; on the nonsymmetric mixed matrix it takes 748 and converges at every tested level. The measured $\delta_A = 0.03954$ at $N = 64$ quantifies the mixed-matrix nonsymmetry, and CG convergence at every tested level is reported as an experimental result for this matrix family. CG and CG+Jacobi both take 221 iterations on the constant-diagonal Dirichlet system at $N = 64$, consistent with diagonal preconditioning amounting to a constant scaling in that case.

4.5 Matched FDM and P1 FEM

The matched pure-Dirichlet benchmark is evaluated at $N = 8, 16, 32, 64$. Physical nodes, interior DOFs, common-node ordering, norms, and direct-solver protocol are fixed before the methods are compared. At all four levels, the matrices have identical nonzero patterns and zero pattern mismatches; the relative discrepancy in the scaled-stencil audit is 1.9984×10^{-15}. This is the numerical counterpart of $h^2 A_{\text{FDM}} = K_{\text{P1}}$ in Equation (20).

A representative matched-cost run at $N = 64$ separates assembly from the linear solve. The times are medians of five local runs with the same direct solver; the error and observed order are the values from the refinement sequence below. The two methods have the same DOF count, graph, and nnz, so the assembly contrast is the meaningful cost difference.

Table 8. Representative matched benchmark at $N = 64$. Assembly and solve times are reported separately; errors and observed orders use the global discrete L_h^2 sequence.

Method	DOFs	Assembly (s)	Solve (s)	nnz	L_h^2 error	Order
FDM	3969	0.00758	0.00600	19593	4.8369×10^{-6}	1.997395
P1 FEM	3969	0.14372	0.00619	19593	9.6741×10^{-6}	1.997659

Table 9. Matched pure-Dirichlet FDM/P1-FEM benchmark from the matched implementation. Fits use $N = 16, 32, 64$; equal nodes and DOFs make the differing error constants directly comparable.

Data	Method	L_h^2 order	L_∞ order	L_h^2 at $N = 64$	L_∞ at $N = 64$
Exponential	FDM	1.997395	1.996838	4.8369×10^{-6}	8.7856×10^{-6}
Exponential	P1 FEM	1.997659	1.997102	9.6741×10^{-6}	1.7572×10^{-5}
Sine	FDM	2.001304	2.001304	1.0041×10^{-4}	2.0082×10^{-4}
Sine	P1 FEM	1.998460	1.998696	1.0210×10^{-4}	2.0077×10^{-4}

Both discretisations retain approximately second-order nodal accuracy in Table 9. Their error constants differ: with exponential data at $N = 64$, the P1 L_h^2 error is approximately twice the FDM value. The matched matrix audit attributes this difference to the load vectors, because b_{P1} contains integrated moments $\int f\phi_i \, dx$ whereas b_{FDM} contains nodal samples. This pure-Dirichlet control complements the mixed-boundary experiments.

4.6 Smooth and Singular P1 Controls

The P1 study applies one solver, one mesh sequence, and one integration protocol to smooth $u_s = e^{x+y}$ and singular $u^* = r^{2/3}\sin(2\phi/3)$ data. Here $h = 2/n$, and both problems use the same DOF sequence. Table 10 reports the continuous errors; Figure 7 displays their refinement trends.

Table 10. Smooth and singular P1 errors on identical L-shaped meshes. The archived global errors use analytic radial integration with angular quadrature, Duffy tensor-Gauss integration away from the corner, and continuous domain norms.

n	DOFs	Smooth L^2	Smooth energy	Singular L^2	Singular energy
8	33	2.7998×10^{-2}	3.8972×10^{-1}	1.8812×10^{-2}	1.9289×10^{-1}
16	161	6.9873×10^{-3}	1.9568×10^{-1}	7.5922×10^{-3}	1.2402×10^{-1}
32	705	1.7459×10^{-3}	9.7945×10^{-2}	3.0235×10^{-3}	7.9193×10^{-2}
64	2945	4.3642×10^{-4}	4.8985×10^{-2}	1.1972×10^{-3}	5.0324×10^{-2}
128	12033	1.0910×10^{-4}	2.4494×10^{-2}	4.7313×10^{-4}	3.1878×10^{-2}

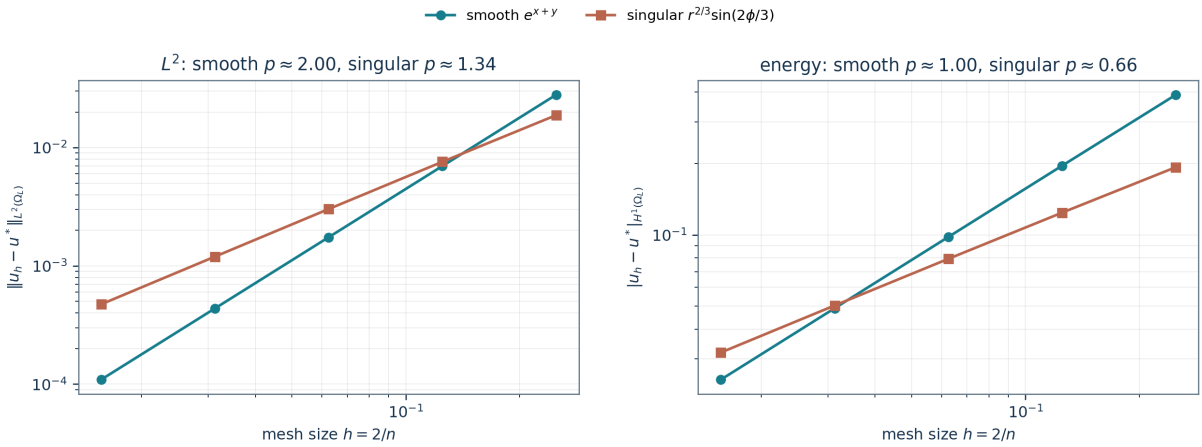


Figure 7. The measured smooth–singular regularity gap in continuous P1 norms. Smooth L^2 and energy orders are approximately 2 and 1; singular orders approach $4/3$ and $2/3$, respectively.

The smooth pairwise L^2 order is 2.00 at every refinement; its energy order is 0.99 on the first refinement and 1.00 thereafter. The singular L^2 orders are 1.31, 1.33, 1.34, and 1.34, while the energy orders are 0.64, 0.65, 0.65, and 0.66. They approach limiting orders $4/3$ in L^2 and $2/3$ in energy, consistent with the sub-endpoint estimates in Equations (23)–(24); the smooth rates are 2 and 1. Expressed against DOFs on these two-dimensional meshes, the singular energy orders range from 0.28 to 0.32, near the expected $N_{\mathrm{dof}}^{-1/3}$ scaling; the finest singular L^2 order against DOFs is 0.66, near $N_{\mathrm{dof}}^{-2/3}$.

The corner-region calculation uses $\Omega_L \cap \{r < 0.25\}$. Independent polar integration gives 53.49%, 60.00%, 62.78%, 64.03%, and 64.52% of the squared L^2 error inside this region for $n = 8, 16, 32, 64, 128$. At $n = 128$, the inside and remainder norms are approximately 3.800×10^{-4} and 2.818×10^{-4}; the square root of their squared sum gives the global norm 4.7313×10^{-4} in Table 10. The remainder is obtained by subtracting the inside squared norm from the global squared norm. Doubling the polar Gauss orders from 256×512 to 512×1024 changes the inside norm by at most 1.8×10^{-5} relative across the sequence. Mesh area remains three.

The quadrature self-convergence test quantifies the effect of integration order. For the smooth leg at $n = 32$, Duffy orders two, three, and eight give $1.66206435 \times 10^{-3}$, $1.74489753 \times 10^{-3}$, and $1.74590784 \times$

10^{-3}. The order-two rule underestimates the production value by about 4.8%; the order-eight value is the one used in the reported convergence sequence.

4.7 Interpolation and Discrete-Error Components

The decomposition in Equation (25) distinguishes interpolation error $\rho = u^* - u_I$ from the discrete component $\theta = u_h - u_I$. Table 11 shows that both decrease at approximately second order on the smooth leg. At $n = 128$, $\|\theta\|_{L^2}$ is only 7.89% of the total error, while $\|\rho\|_{L^2}$ closely tracks it. The total smooth error is dominated by the expected interpolation scale, and the measurements show second-order convergence for the total error and both components.

Table 11. Smooth-leg decomposition $e = \theta - \rho$. The piecewise-affine θ is integrated exactly by an elementwise quadratic rule; ρ and e use the common high-order protocol.

n	$\|e\|_{L^2}$	$\|\theta\|_{L^2}$	$\|\rho\|_{L^2}$
8	2.7998×10^{-2}	1.9943×10^{-3}	2.9288×10^{-2}
16	6.9873×10^{-3}	5.3762×10^{-4}	7.3418×10^{-3}
32	1.7459×10^{-3}	1.3697×10^{-4}	1.8367×10^{-3}
64	4.3642×10^{-4}	3.4397×10^{-5}	4.5925×10^{-4}
128	1.0910×10^{-4}	8.6071×10^{-6}	1.1482×10^{-4}

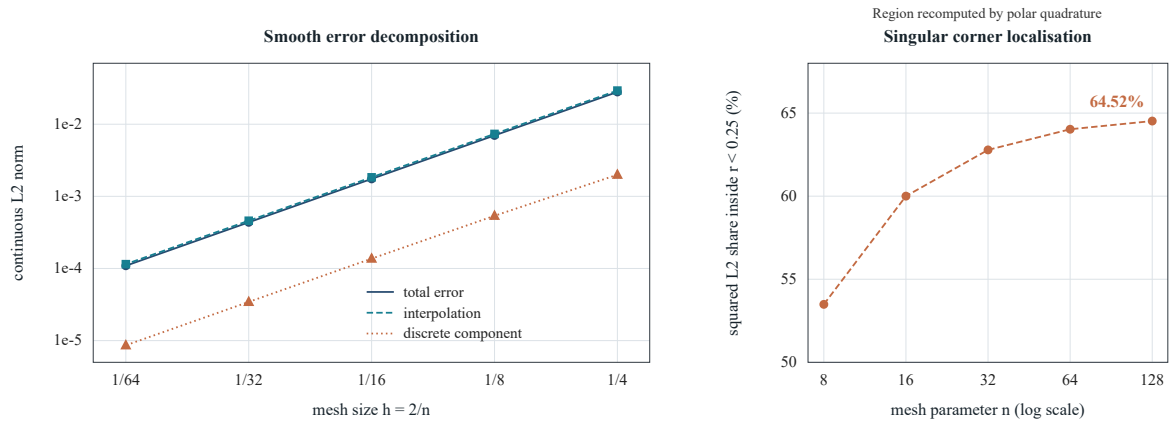


Figure 8. Smooth error decomposition and singular corner localisation. The left panel compares total, interpolation, and discrete components; the right panel shows the share of singular squared L^2 error inside $r < 0.25$.

Figure 8 visualises the measured θ/ρ behaviour alongside the singular corner shares. The smooth components explain the full smooth rate, while the singular corner shares and energy slopes supply independent evidence for the regularity-limited case.

4.8 Structured Grading and Pure-Neumann Audits

The structured radial-grading study compares error against actual DOFs, as appropriate for singular problems and graded meshes [14]. On uniform meshes, fitted energy and L^2 exponents in $E \propto N_{\mathrm{dof}}^{-p}$ are 0.320713 and 0.653727. Radial grading with $\kappa = 6$ raises them to 0.475110 and 0.965005. At outer parameter $N = 32$, the uniform mesh has 705 DOFs and energy error 7.9193×10^{-2}; the graded mesh has 3609 DOFs and energy error 1.8651×10^{-2}. The corresponding L^2 errors are 3.0235×10^{-3} and 1.9757×10^{-4}. The uniform energy baseline is taken from the same Duffy-order-eight sequence reported in Table 10, so the comparison uses one consistent reference run.

The improvement is norm-specific: the nodal L_∞ exponent changes from 0.320422 to 0.335816. Radial grading is prescribed before the solve; estimator-driven adaptivity is a separate algorithmic paradigm [15]. The uniform data and the Duffy-order-eight integration sequence use the same prescribed-family protocol, so the reported outcomes are measured errors and rates for that family.

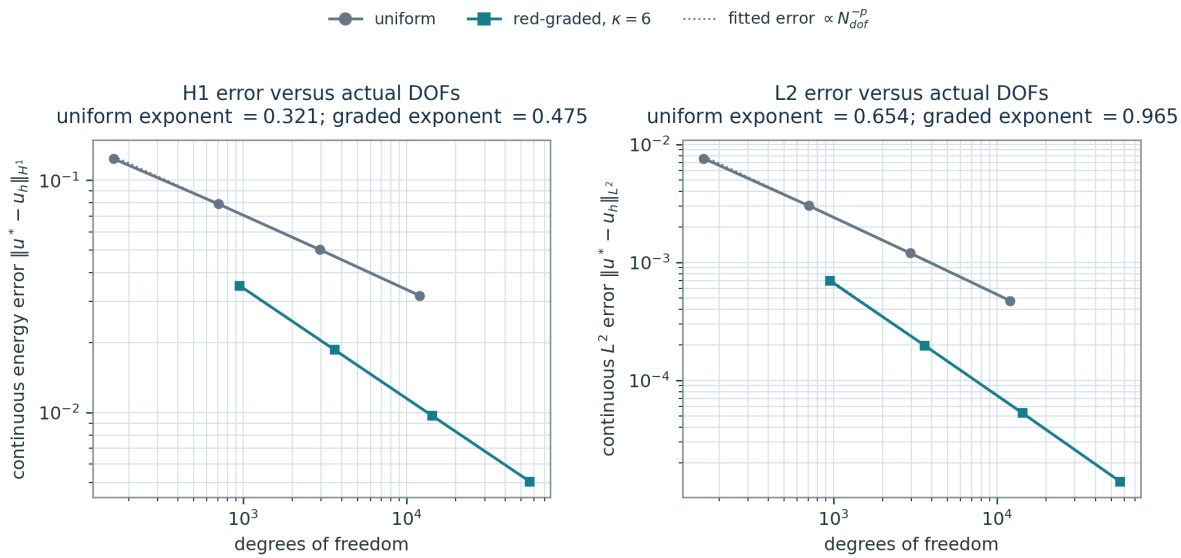


Figure 9. Structured radial grading improves continuous energy and L^2 efficiency against actual DOFs in the structured radial-grading study. The sequence belongs to a prescribed structured mesh family.

A final pure-Neumann audit tests the compatibility and gauge equations developed in Section 2. With quadrature order two, the pre-correction load defect falls from 1.862427×10^{-5} at $N = 4$ to 4.562260×10^{-9} at $N = 32$, approximately by sixteen per mesh halving. After the correction in Equation (28), absolute defects are at most 4.4409×10^{-16}. The mean-gauged solution has zero displayed mean error relative to $(e-1)^2$, and small dense spectra at $N = 4, 8$ each contain exactly one eigenvalue below 10^{-10}. The last-three-level continuous L^2 fit is 1.984688 with order-two quadrature and 1.984610 with order-eight quadrature.

5 Conclusions and Recommendations

5.1 Mixed-Boundary Closure

Across five refinement levels, one-sided Neumann and Robin closures approach first-order global accuracy, whereas ghost elimination approaches second order in global, interior, maximum, and right-edge norms. The discrete Green relation $e = -A^{-1}\tau$ explains why a complete ghost row can have an unscaled $O(h)$ local defect while the global solution converges at second order. Ghost elimination is therefore the preferred closure for the aligned square grids and smooth data studied here.

The verification chain combines exact differentiation of f, q, r, independent row substitution, direct sparse solves, and regional norms. This chain assigns observed differences to the intended numerical mechanism: the closure controls the asymptotic order, while the direct residual remains at 10^{-14} and the independent ordering and P1 controls reproduce the same trends.

5.2 Solver Tolerances and Cost

The GMRES sweep makes algebraic error an explicit budget. At $N = 128$, rtol 10^{-8} produces a maximum error 1.757 times the direct reference; rtol 10^{-10} reaches the discretisation plateau within 0.9%, and rtol 10^{-12} gives error norms within 0.02% of the direct-reference values. Agreement of these error norms alone does not measure the difference between the two solution vectors. For the ghost Neumann M-matrix, the bound $\|x - x_h^*\|_\infty \le B = 2N^2\rho_{\max}$ controls the error relative to the exact discrete solution x_h^*. Using the observed maximum error $E_{\text{obs},\infty}$ gives $R_{\text{cert}} = B/(E_{\text{obs},\infty} - B) < 0.078 < 0.1$ at $N = 128$ and rtol 10^{-10}. This is a bound from the archived CSV residuals, rather than a measured solution difference or a fresh GMRES run. SuperLU is fastest for every tested size; GMRES+ILU reaches the fewest iterations, but ILU construction dominates its total cost.

5.3 Regularity, Efficiency, and Scope

On identical L-shaped meshes, smooth P1 errors approach $(2, 1)$ in (L^2, H^1), while the $r^{2/3}$ singular solution approaches $(4/3, 2/3)$. The θ/ρ decomposition confirms second-order interpolation and discrete components, and 53.49%–64.52% of the singular squared L^2 error lies inside $r < 0.25$, reaching 64.52% at $n = 128$. Independent recomputation reproduces the global L^2 errors for the singular solution in Table 10, while polar quadrature verifies the corner-region contributions. The structured-grading comparison is presented in Section 4.8.

The resulting recommendations are specific: use ghost elimination for aligned mixed-boundary grids, set iterative tolerances from the algebraic/discretisation split, and report total setup-plus-iteration cost when comparing solvers. The next study should extend the same evidence chain to variable coefficients, thin boundary layers, and adaptive treatment of re-entrant corners.

References

[1] LeVeque, R. J. (2007). *Finite difference methods for ordinary and partial differential equations: Steady-state and time-dependent problems*. Society for Industrial and Applied Mathematics. https://doi.org/10.1137/1.9780898717839

[2] Gibou, F., Fedkiw, R. P., Cheng, L.-T., & Kang, M. (2002). A second-order-accurate symmetric discretization of the Poisson equation on irregular domains. *Journal of Computational Physics, 176*(1), 205–227. https://doi.org/10.1006/jcph.2001.6977

[3] Roache, P. J. (1998). Verification of codes and calculations. *AIAA Journal, 36*(5), 696–702. https://doi.org/10.2514/2.457

[4] Roy, C. J. (2005). Review of code and solution verification procedures for computational simulation. *Journal of Computational Physics, 205*(1), 131–156. https://doi.org/10.1016/j.jcp.2004.10.036

[5] Salari, K., & Knupp, P. (2000). *Code verification by the method of manufactured solutions* (SAND2000-1444). Sandia National Laboratories. https://doi.org/10.2172/759450

[6] Saad, Y., & Schultz, M. H. (1986). GMRES: A generalized minimal residual algorithm for solving nonsymmetric linear systems. *SIAM Journal on Scientific and Statistical Computing, 7*(3), 856–869. https://doi.org/10.1137/0907058

[7] Li, X. S. (2005). An overview of SuperLU. *ACM Transactions on Mathematical Software, 31*(3), 302–325. https://doi.org/10.1145/1089014.1089017

[8] Grisvard, P. (2011). *Elliptic problems in nonsmooth domains*. Society for Industrial and Applied Mathematics. https://doi.org/10.1137/1.9781611972030

[9] Brenner, S. C., & Scott, L. R. (2008). *The mathematical theory of finite element methods* (3rd ed., Vol. 15). Springer. https://doi.org/10.1007/978-0-387-75934-0

[10] Gustafsson, T., & McBain, G. D. (2020). scikit-fem: A Python package for finite element assembly. *Journal of Open Source Software, 5*(52), 2369. https://doi.org/10.21105/joss.02369

[11] Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature, 585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

[12] Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., … SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods, 17*(3), 261–272. https://doi.org/10.1038/s41592-019-0686-2

[13] Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering, 9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55

[14] Apel, T., Sändig, A.-M., & Whiteman, J. R. (1996). Graded mesh refinement and error estimates for finite element solutions of elliptic boundary value problems in non-smooth domains. *Mathematical Methods in the Applied Sciences, 19*(1), 63–85. https://doi.org/10.1002/(SICI)1099-1476(19960110)19:1<63::AID-MMA764>3.0.CO;2-S

[15] Dörfler, W. (1996). A convergent adaptive algorithm for Poisson's equation. *SIAM Journal on Numerical Analysis, 33*(3), 1106–1124. https://doi.org/10.1137/0733054

Appendix

A Team AI Transparency Log

This appendix records the project's account of AI use and the numerical claim checks made during this report revision. Routine spelling, formatting, and language suggestions are grouped in A1; A2–A3 identify the specific claims corrected in the revision and the evidence used. These revision checks do not establish which checks individual team members performed originally. Raw transcripts are not reproduced here.

A1. AI tools used

Tool or source	Period	Primary use
AI assistants listed in the retained logs (Kimi/Moonshot AI, TeleAgent, ChatGPT/Codex)	Sep.–Oct. 2026	Drafting and reviewing the mathematical model, boundary closures, solver and FEM experiments, audit checklists, figures, tables, and report prose. AI output is treated as a proposal. The revision checks the supplied report source and CSV evidence and states the limits imposed by missing iterate vectors.

A2. Significant AI-assisted claim audits

Item	Prompt, claim, and assumptions	Risk and possible falsifier
1	The row-defect prose named $y = 0.37$, whereas the audit values correspond to the shared grid node $y = 3/8$. The local ghost-row defect is $O(h)$ while the global solution remains second order.	A coordinate mismatch or confusion between a row defect and a solution error could misstate the evidence. Check the tabulated defects at the actual shared node and fit global error norms independently.
2	The solver prose claimed that rtol 10^{-12} reproduced the displayed direct-reference digits and described forward-error and stagnation gates.	The archive contains CSV diagnostics rather than iterate vectors, so aggregate error-norm agreement cannot verify a measured solution difference. Check the relative error-norm differences and certify the algebraic budget using the ghost Neumann M-matrix bound.
3	The corner-region prose attributed about 92% of the singular squared L^2 error to $r < 0.25$.	An incorrect regional share can misrepresent error localisation even when global norms are correct. Reintegrate independently in polar coordinates and compare both the global norm and the corner-to-total squared-error ratio.

A3. Independent verification and decision

Item	Independent check and evidence location	Final decision, modification, or limitation
1	The revision checks the row-defect values at the actual common node $y = 3/8$ against the complete-row formulas and the refinement data in Sections 2.3 and 4.2.	The coordinate labels are corrected to $y = 3/8$. Local defect and global solution error remain separate diagnostics, and order claims are restricted to the tested grids and data.
2	The revision compares the archived tolerance-sweep error norms with the direct reference and derives $\|x - x_h^*\|_\infty \le B = 2N^2\rho_{\max}$ for the exact ghost Neumann discrete solution x_h^*. At $N = 128$ and rtol 10^{-10}, the certified algebraic-to-discretisation ratio is $R_{\text{cert}} = B/(E_{\text{obs},\infty} - B) < 0.078$.	Replace displayed-digit agreement by error-norm agreement within 0.02% at rtol 10^{-12}. The bound meets the 0.1 budget at rtol 10^{-10}; missing iterate vectors prevent a measured solution-difference or successive-iterate check. No fresh GMRES runs are claimed.
3	Independent recomputation reproduces the global L^2 errors for the singular solution in Table 10; polar quadrature verifies the corner-region contributions, giving squared-error shares of 53.49%–64.52%, with 64.52% at $n = 128$.	Replace the unsupported 92% localisation claim by these shares for $r < 0.25$. The original structured-grading comparison is retained in Section 4.8; this editorial restoration does not claim a fresh graded-mesh solve.

A4. Intellectual ownership declaration

The team owns the project-specific grid and indexing, boundary rows, solver updates, residual and error metrics, exact-row verification, singular-corner integration, pure-Neumann compatibility and gauge, E1–E6 experiment logic, figures, and report text. Sparse storage and mature factorisations are delegated to established libraries; the checks and limitations in A3 specify the evidence available for the revised numerical claims. Each member affirms that they can explain and modify the team-owned numerical methods and justify the assumptions used at those interfaces without AI assistance during the viva.

Name (ID)	Team-owned artefact or contribution	Trusted dependency and interface
Yi Zhu (2361666)	Project coordination, task allocation, requirements checks, and final report/submission integration.	Git/repository and course-submission interfaces.
Zhengdao Song (2361511)	Continuous model, discrete analysis, boundary closures, error metrics, solver theory, pure-Neumann and L-shaped-domain theory.	Hand derivations and cited theory; no unexamined solver delegation.
Xiangyu Zhang (2361653)	Implementation modules, E1–E6 drivers, reproducibility workflow, figures, tables, and report source.	NumPy/SciPy, SuperLU/GMRES, scikit-fem, quadrature, and Matplotlib interfaces.
Fangchi Shi (2359837)	Algorithm workflow implementation support, indexing and boundary-row checks, solver stopping criteria, and output diagnostics.	NumPy/SciPy solver interfaces and the team's exact-row and refinement-test protocols.
Rong Yin (2363466)	Independent reference-solution comparisons, testing, validation records, and error-integration audits.	Direct and iterative solver interfaces, manufactured-data checks, and the validation test suite.

B Individual Contribution Statements

Each statement below follows the four-question format used in the reference report. The descriptions are intentionally concise but traceable to the report sections, experiment drivers, and validation records.

B1. Yi Zhu (2361666) – Project Manager

1. Specific contributions.

1. Prepared structured learning notes from the course material and reorganised the main ideas into a sequence that the team could use during development, revision, and the final viva preparation.
2. Created individual task lists with clear requirement links, ownership, deliverables, and review points, so that each member could trace a task from the project brief to a report section or experiment.
3. Coordinated the mathematical, implementation, and validation work through weekly milestones, followed up dependencies between members, and kept the separate evidence streams aligned.
4. Reviewed the report against the project brief, checked mandatory figures, tables, and explanations, identified consistency problems, and coordinated the final revisions, packaging, and submission materials.

2. Hardest conceptual obstacle. Converting a broad project brief into a complete evidence chain with clear owners and review dates, while keeping mathematical predictions, numerical experiments, and report claims aligned.

3. AI output used and what changed. A report-review and checklist draft was used to locate omissions and possible inconsistencies. The team did not accept its conclusions as evidence: the relevant experiments, tables, and page-level checks were completed before the final wording was retained.

4. One thing to improve. I would establish a shared checklist with an owner and a review date for every requirement at the beginning, and seek earlier feedback on the learning notes and task lists.

Signature: 朱燈 Date: 10.31

B2. Zhengdao Song (2361511) – Mathematical Theory

1. Specific contributions.

1. Defined the mixed Poisson model, manufactured data, weak forms, boundary ownership, and well-posedness assumptions in Sections 1.1 and 2.1.
2. Derived the five-point stencil, Dirichlet elimination, one-sided and ghost Neumann/Robin rows, their local defects, and the error and residual metrics in Sections 2.2–2.4.
3. Supplied the matrix and solver rationale, pure-Neumann compatibility and mean gauge, and the L-shaped regularity theory, including the $(4/3, 2/3)$ prediction and the paired smooth/singular protocol.

2. Hardest conceptual obstacle. Separating a complete-row defect from the global solution error, and explaining how a locally first-order boundary defect can coexist with a second-order ghost solution through the discrete Green relation.

3. AI output used and what changed. AI-assisted derivations were treated as drafts. I re-derived the signs, factors of h, compatibility projection, gauge condition, corner branch convention, and regularity estimates by hand; an early prediction of extra P1 diagonal couplings was rejected after the assembly audit.

4. One thing to improve. I would test structural matrix claims on a small mesh before releasing the modelling document and specify the graded-mesh mapping and parameter at the prediction stage.

Signature: 宋政道 Date: 10.31

B3. Xiangyu Zhang (2361653) – Algorithm Implementation; Visualization and Report

1. Specific contributions.

1. Implemented the data and indexing layers in `data.py`, including the $N(N-1)$ unknown map, sparse assembly, strong Dirichlet elimination, and one-sided and ghost Neumann/Robin closures.
2. Implemented the direct, stationary, and GMRES/ILU solver routines and centralised row scaling, residual metrics, error norms, pairwise orders, and energy-error calculations so that all experiments used the same definitions.
3. Built the E1–E2 solver-accuracy, tolerance, stagnation, SOR-scan, and cost experiments, including the forward-error gate and the separate common-residual and matched-accuracy timing records.
4. Constructed the matched FDM/P1-FEM benchmark, aligned physical coordinates and degrees of freedom, and added coordinate-order, zero-pattern, and scaled-stencil assertions to separate matrix equivalence from load-vector effects.
5. Implemented the E4–E6 geometry, regularity, pure-Neumann compatibility, and mean-gauge diagnostics, including the analytic radial/adaptive-angular corner integrator and the prescribed graded-mesh comparisons.
6. Maintained the project's `run_all.py` workflow, generated figures and tables through the shared plotting utilities, and wrote and typeset the implementation, results, algorithms, captions, and conclusions with page-level consistency checks.

2. Hardest conceptual obstacle. Keeping implementation order, physical coordinates, error definitions, and report tables consistent across six experiment drivers, especially when a plausible plot could conceal an indexing or corner-integration error.

3. AI output used and what changed. AI-assisted implementation and experiment suggestions were treated as drafts. The checks supported by the supplied archive are described in Section 3.1 and Appendix A3. The original `run_all.py` and `validation/` files are absent, so clean reproduction of the full workflow and the original forward-error and stagnation gates cannot be verified from this archive.

4. One thing to improve. I would freeze a single results schema and the dual accuracy/stagnation gate before the first full run, and request an independent review of singular-corner quadrature and table-to-CSV mapping earlier.

Signature: 张祥宇. Date: 2026.10.31

B4. Fangchi Shi (2359837) – Algorithm Implementation

1. Specific contributions.

1. Supported the implementation of the three reported algorithm workflows and checked that their indexing, boundary-row updates, stopping criteria, and output diagnostics matched the mathematical specification.

2. Hardest conceptual obstacle. Keeping the boundary-row scaling and solver stopping rules consistent when the same mixed system was passed through direct, stationary, and Krylov solver paths.

3. AI output used and what changed. An AI-generated pseudocode draft was rewritten to match the team's actual data structures and algorithm interfaces; the boundary signs and update order were then checked against exact-row substitution and the reported refinement tables.

4. One thing to improve. I would request an independent review of the solver interface and a small end-to-end benchmark earlier, before integrating the algorithm descriptions into the final report.

Signature: 施方驰 Date: 2026.10.31

B5. Rong Yin (2363466) – Testing and Validation

1. Specific contributions.

1. Built an independent manufactured-data FDM reference solver and verified forcing, boundary signs, corner ownership, all four right-edge rows, scaled residuals, and the four closure/condition convergence sequences.
2. Built the independent P1 FEM validation branch on exactly represented L-shaped meshes, including the area check, corner integration, energy errors, and head-to-head comparison with the implementation branch.
3. Ran the tolerance and iterative-method audits, documented quadrature and free-corner retractions, and maintained the evidence records and final validation checklist.

2. Hardest conceptual obstacle. A small algebraic residual did not guarantee a correct boundary discretisation, and an apparently stable quadrature rule produced a misleading L2 trend. Independent row substitution, corner constraints, and cross-branch comparison were needed to expose both issues.

3. AI output used and what changed. AI-generated solver and audit suggestions were subjected to a five-gate review. I corrected the ghost-row sign before publication, replaced weak corner quadrature with analytic radial/adaptive angular integration, and retained only numbers reproduced by the independent validation branch.

4. One thing to improve. I would budget the boundary-node and corner-integrator tests before the first error table and request a callable cross-check interface for the main solver earlier.

Signature: 尹榕 Date: 2026.10.31